Spatial Transcriptomics Cheatsheet

The tables below consist of valuable functions or commands that will help you through this module.

Each table represents a different library/tool and its corresponding commands.

> Please note that these tables are not intended to tell you all the information you need to know about each command.
>
> The hyperlinks found in each piece of code will take you to the documentation for further information on the usage of each command. Please be aware that the documentation will generally provide information about the given function's most current version (or a recent version, depending on how often the documentation site is updated). This will usually (but not always!) match what you have installed on your machine. If you have a different version of R or other R packages, the documentation may differ from what you have installed.

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`SpatialExperiment`

Read the `SpatialExperiment` package documentation (and e-book) and a vignette on its usage.

`SpatialExperiment` objects are based on `SingleCellExperiment` objects. There are many functions that are shared between the two, with equivalent usage. See this `SingleCellExperiment` vignette for example usage of shared functions. You can also refer to the Data Lab cheatsheets for our "Introduction to scRNA-seq" (`scRNA-seq-cheatsheet.pdf`) and "Advanced scRNA-seq" (`scRNA-seq-advanced-cheatsheet.pdf`) training modules.

Library/Package	Piece of Code	What it's called	What it does
`SpatialExperiment`	`SpatialExperiment()`	Spatial Experiment	Creates a `SpatialExperiment` object
`SpatialExperiment`	`spatialCoords()`	Spatial Experiment coordinates	Extracts spatial coordinates of spots in the `SpatialExperiment` object
`SpatialExperiment`	`imgData()`	Spatial Experiment images	View the image data associated with a `SpatialExperiment` object

`VisiumIO`

Read the R package `VisiumIO` documentation and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`VisiumIO`	`TENxVisium()`	10x Visium	Constructor function for importing 10x Visium data
`VisiumIO`	`import()`	Import	Loads 10x Visium data defined with the `TENxVisium()` constructor function

`ggspavis`

Read the `ggspavis` package documentation and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`ggspavis`	`plotCoords()`	Plot coordinates	Creates a spot plot showing spatial locations in x/y with options for spot annotation
`ggspavis`	`plotVisium()`	Plot Visium	Creates a spot plot for Visium data specifically showing spatial locations in x/y with options for spot annotation as well as an option to show the spatial image, e.g., the H&E image

`SpotSweeper`

Read the `SpotSweeper` package documentation and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`SpotSweeper`	`localOutliers()`	Local outliers	Detect local outlier spots in a `SpatialExperiment` based on a given quality control metric
`SpotSweeper`	`plotQCmetrics()`	Plot QC metrics	Create a spot plot highlighting outliers detected with `SpotSweeper::localOutliers()`

`Banksy`

Read the `Banksy` package documentation, which contains several vignettes on its usage, including a vignette on parameter selection. Note that many of their vignettes use of image-based spatial transcriptomics data that is more fine-grained compared to 10x Visium.

Library/Package	Piece of Code	What it's called	What it does
`Banksy`	`computeBanksy()`	Compute `Banksy`	Compute the component neighborhood matrices for the `Banksy` matrix
`Banksy`	`runBanksyPCA()`	Run `Banksy` PCA	Run PCA on a `Banksy` matrix with a weighted spatial neighborhood context

`spacexr`

Read the `spacexr` package documentation, and a vignette on using their RCTD algorithm.

Library/Package	Piece of Code	What it's called	What it does
`spacexr`	`createRctd()`	Create an RCTD object	Preprocess data before performing deconvolution with RCTD
`spacexr`	`runRCTD()`	Run RCTD	Run the RCTD algorithm to decompose cell type mixtures
`spacexr`	`plotCellTypeWeight()`	Plot a single cell type's weights	Plot pixel proportions for a specific cell type on a spatial coordinates plot

`SpatialExperimentIO`

Read the `SpatialExperimentIO` package documentation and a vignette on its usage.

This package provides reader functions for imaging-based spatial transcriptomics platforms, including Xenium (10x Genomics), CosMx (NanoString), MERSCOPE (Vizgen), and STARmap PLUS.

Library/Package	Piece of Code	What it's called	What it does
`SpatialExperimentIO`	`readXeniumSXE()`	Read Xenium into a Spatial Experiment	Loads an unzipped 10x Genomics Xenium output bundle directory into a `SpatialExperiment` object

`hoodscanR`

Read the `hoodscanR` package documentation and a vignette on its usage.

Note that the functions in this package are generally run in sequence, where the output of one function is the input to the next.

Library/Package	Piece of Code	What it's called	What it does
`hoodscanR`	`findNearCells()`	Find the k-the nearest cells for each	Find the `k` nearest cells to each cell in a `SpatialExperiment` object, returning a list with a matrix of distances and a matrix of the annotations (e.g., cell types) of those neighboring cells
`hoodscanR`	`scanHoods()`	Scan cellular neighborhoods	Apply a modified softmax algorithm to the distance matrix from `hoodscanR::findNearCells()` to calculate the probability that each cell is associated with each of its nearest neighboring cells
`hoodscanR`	`mergeByGroup()`	Merge probability matrix based on annotations	Collapse the probability matrix from `hoodscanR::scanHoods()` by annotation, so that each column is a group (e.g., a cell type) rather than an individual neighboring cell
`hoodscanR`	`mergeHoodSpe()`	Merge probability matrix into Spatial Experiment object	Add the merged probability matrix into the `colData` of a `SpatialExperiment` object
`hoodscanR`	`calcMetrics()`	Calculate metrics for probability matrix	Calculate the entropy and perplexity of the probability matrix for each cell, which summarize how mixed or how distinct that cell's neighborhood is, and store them in `colData`
`hoodscanR`	`plotColocal()`	Plot heatmap for neighborhood analysis	Plot a heatmap of the Pearson correlations between cells' neighborhood probability distributions to show which groups tend to colocalize
`hoodscanR`	`clustByHood()`	Cluster the probability matrix with K-means	Perform k-means clustering on the probability matrix to assign each cell to a neighborhood cluster
`hoodscanR`	`plotProbDist()`	Plot probability distribution	Plot the distribution of neighborhood probabilities, optionally split by a grouping variable such as the neighborhood clusters from `hoodscanR::clustByHood()`

`scuttle`, `scran`, and `scater`

Read the `scuttle` package documentation, and a vignette on its usage.

Read the `scran` package documentation, and a vignette on its usage.

Read the `scater` package documentation, and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`scuttle`	`addPerCellQC()`	Add per cell quality control	For a `SingleCellExperiment` object, calculate and add quality control per cell and store in `colData`
`scuttle`	`computeLibraryFactors()`	Compute Library Factors	Returns a numeric vector of computed size factors for each spot (or cell) stored in a `SpatialExperiment` (or `SingleCellExperiment`) object. The size factor is computed as the library size of each spot/cell after scaling them to have a mean of 1 across all spots/cells
`scuttle`	`logNormCounts()`	Normalize log counts	Returns the `SpatialExperiment` (or `SingleCellExperiment`) object with normalized expression values for each spot (cell), using the size factors stored in the object
`scuttle`	`isOutlier()`	Identify outliers	Convenience function to determine which values in a numeric vector are outliers based on the median absolute deviation (MAD)
`scran`	`getTopHVGs()`	Get top highly variable genes	Identify variable genes in a `SingleCellExperiment` object, based on variance
`scran`	`modelGeneVar()`	model per gene variance	Model the per gene variance of a `SingleCellExperiment` object
`scran`	`clusterCells()`	Cluster cells	Perform clustering on an SCE object using the `bluster` package
`scater`	`runPCA()`	Run PCA	Calculates principal components analysis on a `SingleCellExperiment` object, returning an SCE object with a PCA reduced dimension
`scater`	`runUMAP()`	Run UMAP	Calculates uniform manifold approximate projection on a `SingleCellExperiment` object, returning an SCE object with a UMAP reduced dimension
`scater`	`plotUMAP()`	Plot UMAP	Plot the "UMAP"-named reduced dimension slot from a `SingleCellExperiment` object

`bluster`

Read the `bluster` package documentation and vignettes on its usage:

- Flexible clustering for Bioconductor
- Assorted clustering diagnostics

Library/Package	Piece of Code	What it's called	What it does
`bluster`	`NNGraphParam()`	Graph-based clustering parameters	Set up parameters for nearest-neighbor (NN) graph-based clustering algorithms within `scran::clusterCells()` or `bluster::clusterRows()`
`bluster`	`approxSilhouette()`	Approximate silhouette width	Calculate an approximate silhouette width for each cell given a set of clusters
`bluster`	`neighborPurity()`	Compute neighborhood purity	Calculate neighborhood purity for each cell given a set of clusters
`bluster`	`bootstrapStability()`	Assess cluster stability by bootstrapping	Generate cluster bootstrap replicates to estimate cluster robustness to sampling noise
`bluster`	`pairwiseRand()`	Compute pairwise Rand indices	Calculate the Rand index (adjusted, by default) between pairs of clustering results, using one of several available calculation modes

`patchwork`

Read the `patchwork` package documentation, and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`patchwork`	`wrap_plots()`	Wrap plots	Wrap multiple `ggplot2` objects into a single multi-panel plot
`patchwork`	`+`	Plot arithmetic	Place `ggplot2` objects side-by-side by adding them together with a `+`. The `patchwork` package must be loaded to use this symbol with plots
`patchwork`	`/`	Plot arithmetic	Stack `ggplot2` objects on top of one another "dividing" them with a `/`. The `patchwork` package must be loaded to use this symbol with plots

`pals`

Read the `pals` package documentation, and a vignette on its usage.

Library/Package	Piece of Code	What it's called	What it does
`pals`	`alphabet()`	Alphabet palette	Create a discrete palette from the "alphabet" set of colors

`pheatmap`

Read the `pheatmap` package documentation.

Library/Package	Piece of Code	What it's called	What it does
`pheatmap`	`pheatmap()`	Pretty heatmap	Plot a (pretty!) clustered heatmap

`purrr`

Read the `purrr` package documentation and a vignette on its usage, and download the `purr` package cheatsheet.

Library/Package	Piece of Code	What it's called	What it does
`purrr`	`map()`	map	Apply a function across each element of list; return a list
`purrr`	`imap()`	imap	Apply a function across each element of list and its index/names; return a list
`purrr`	`reduce()`	Reduce	Reduce a list to a single value by repeatedly applying a given function. Can also be used to iteratively modify a single object.

`ComplexHeatmap`

Read the `ComplexHeatmap` package documentation and the `ComplexHeatmap` Complete Reference e-book.

Library/Package	Piece of Code	What it's called	What it does
`ComplexHeatmap`	`Heatmap()`	Heatmap	Create a clustered heatmap from a matrix, with options to split, annotate, and combine it with other heatmaps